

indication of the conferencing.

The working of the circuit is simple. To check if the wiring of switch S4 is correct, connect the telephone set to line 1. Now lift up the handset and dial the number of line 2. The ringer would sound. Now disconnect line 1 and connect line 2 through switch S4. You would get the dial tone from line 2.

To check a conference call, you would need the help of two friends. First connect switch S4 to line 1 and make a call to friend 1. Now flip the DPST switch S3 to the 'on' position. This puts on hold friend 1 on line 1 and the conference LED3 lights up. Connect switch S4 to line 2 and dial friend 2. When the call on line 2 is answered, a three-way conversation can be made.

When the duophone is not in use,

connect switch S4 to line 1. All other switches should be in the 'off' mode and all LEDs should be unlit. This permits the telephone ringer to be activated if a call comes on line 2. For making calls using line 1 or line 2, you can simply connect switch S4 to the desired line.

Assemble the circuit on a general purpose PCB and enclose it in a suitable cabinet. Fix the switches S1 through S4 on the front side of the cabinet. Also fix the LEDs on the front of the cabinet and the buzzer at the back of the cabinet. It would be better if you use telephone sockets for the telephone lines. Sockets are relatively inexpensive and save time when troubleshooting needs to be done. Use modular plugs to connect the circuit and the two telephone lines. By using such 'quick discon-

nect' plugs, you can easily remove the unit from the telephone lines. Check the polarity of the telephone lines with a multimeter and connect it to the circuit accordingly.

To check the circuit after completing the wiring, connect a 6V regulated power supply to line 1. When you switch S1 to the 'on' position, LED1 blinks at a rate of 2 Hz. If you flip switch S1 to the 'off' position and switch S3 to the 'on' position, LED1 stops blinking and LED3 starts glowing, indicating that the conferencing facility is being used. Now disconnect line 1 from the 6V power supply, connect it to line 2 and flip switch S2 to the 'on' position. Now LED2 blinks at a rate of 2 Hz. Before connecting the circuit to the telephone lines, flip each hold switch to the 'off' position. Now your circuit is ready to be used. ●

MINI UPS SYSTEM

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This circuit provides an uninterrupted power supply (UPS) to operate 12V, 9V and 5V DC-powered instruments at up to 1A current. The backup battery takes up the load without spikes or delay when the mains power gets interrupted. It can also be used as a workbench power supply that provides 12V, 9V and 5V operating voltages. The circuit immediately disconnects the load when

the battery voltage reduces to 10.5V to prevent deep discharge of the battery. LED1 indication is provided to show the full charge voltage level of the battery. Miniature white LEDs (LED2 and LED3) are used as emergency lamps during power failure at night.

A standard step-down transformer provides 12V of AC, which is rectified by diodes D1 and D2. Capacitor C1 provides ripple-free DC to charge the battery and to the remaining circuit. When the mains power is

on, diode D3 gets forward biased to charge the battery. Resistor R1 limits the charging current. Potentiometer VR1 (10K) with transistor T1 acts as the voltage comparator to indicate the voltage level. VR1 is so adjusted that LED1 is in the 'off' mode. When the battery is fully charged, LED1 glows indicating a full voltage level of 12V.

When the mains power fails,

